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RESOURCE ARTICLE

Bayesian hybrid index and genomic cline estimation with the R package *gghybrid*

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Abstract

Admixture, the interbreeding of individuals from differentiated source populations, is now known to be a widespread phenomenon. Genomic studies of natural hybridisation can help to answer many questions on the impacts of admixture on adaptive evolution, reproductive isolation, and speciation. When a large variety of admixture proportions between two source populations exist, both geographic and genomic cline analysis are suitable methods for inferring biased, restricted or excessive gene flow at individual loci into the foreign genomic background, providing evidence for reproductive isolation, selection across an environmental transition, balancing selection, and adaptive introgression. Genomic cline analysis replaces geographic location with genome-wide hybrid index and is therefore useable in circumstances that violate geographic cline assumptions. Here, I introduce *gghybrid*, an R package for simple and flexible Bayesian estimation of Buerkle's hybrid index and Fitzpatrick's logit-logistic genomic clines using bi-allelic data, suitable for both small and large datasets. *gghybrid* allows any ploidy and uses *Structure* input file format. It has separate functions for hybrid index and cline estimation, treating each individual and locus respectively as an independent analysis, making it highly parallelisable. Admixture proportions from other software can alternatively be used in cline analysis, alongside parental allele frequencies. Parameters can be fixed and samples pooled for statistical model comparison with AIC or waic. Here, I describe the functions, pipeline, and statistical properties of *gghybrid*. Simulations reveal that model comparison with waic is preferred, and use of Bayesian posterior distributions and *p* values to select candidate non-null loci is problematic and should be avoided.

KEYWORDS
adaptive introgression, admixture, hybrid zone, reproductive isolation, speciation

1 | INTRODUCTION

Both historical and contemporary exchange of genomic material among genetically differentiated populations, including different species, are now known to be widespread and to have major impacts on evolution, adaptation, and speciation (e.g. Seehausen et al., 2014; Martin & Jiggins, 2017). For example, some modern human adaptations originated from either Neanderthals or Denisovans (Dannemann & Racimo, 2018), rampant adaptive introgression of wing colour genes has occurred across *Heliconius* butterfly species (Edelman et al., 2019), and admixture is thought to have played a defining role in the explosive adaptive radiation of ancient lake cichlid fishes (Meier et al., 2017). The continuing development of tools to examine hybrid evolution is therefore essential.

Admixed populations often harbour polymorphism at loci at or near fixation in non-hybrid populations, allowing studies of reproductive

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incompatibilities (Maheshwari & Barbash, 2011), and the quantitative genetics of species-informative traits. Experimental studies are useful (e.g. Patton et al., 2021; Pereira et al., 2021), however there are many benefits to observational studies of 'natural experiments' such as hybrid zones (Harrison, 1993), and numerous taxa are not amenable to experimentation. In situations whereby (a) a large variety of recombinant types exist allowing genomic regions to evolve independently, and (b) a large variety of admixture proportions between two source populations also exist, cline analysis can be used to detect loci of interest (Barton & Hewitt, 1985; Szymura & Barton, 1986; Gompert & Buerkle, 2011; Fitzpatrick, 2013; Derryberry et al., 2014; Figure 1). While related methods focus primarily on adaptive introgression and hence biased gene flow (e.g. Dias-Alves et al., 2018; Setter et al., 2020), cline analysis can identify restricted, biased, and extreme bi-directional introgression in a single framework.

Geographic cline analysis (Figure 1d) has a foundation in population genetic theory primarily developed by Barton and colleagues (Barton, 1979, 1983; Barton & Gale, 1993; Barton & Hewitt, 1989; Kruuk et al., 1999; Nürnberger et al., 1995). For a locus under direct selection or a neutral locus linked to a single selected locus, the simplest geographic cline function produces a logistic-shaped 'tanhyperbolic' (tanh) curve of changing allele frequency or phenotypic trait value with distance along a one-dimensional geographical transect running perpendicular to the centre of a hybrid zone. Its two parameters, cline *width* and *centre*, respectively indicate the amount of introgression (narrower cline *width*s indicate stronger selection against gene exchange between differentiated populations) and its location (*centre* represents the position along the transect of the maximum slope of the curve, which is also the halfway point between parental values). Estimates of the strength of selection

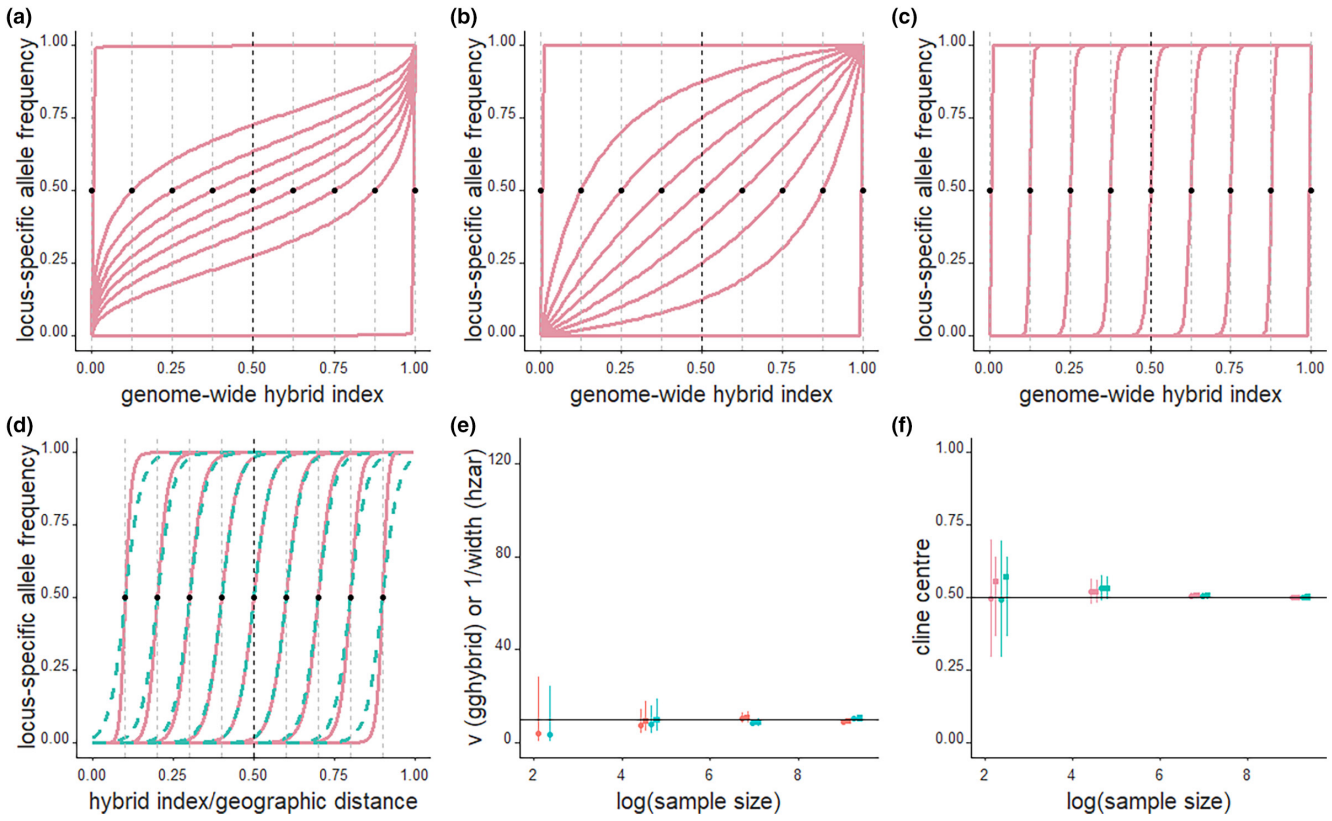


FIGURE 1 Geographic and genomic cline shapes and cline fit comparison. (a–c): Examples of Fitzpatrick's logit-logistic genomic cline curves with varying steepness (*v*) and *centre* (marked with black circles, and a black dashed vertical line for the null value *centre* = 0.5, otherwise grey). Each panel consists of cline curves for 9 evenly spaced *centres* from 0.000001 to 0.999999. (a) *v* = 0.5, a shallower change in locus-specific allele frequency than the genome-wide average, indicating excess bi-directional introgression and suggestive of balancing selection or heterozygote advantage. (b) *v* = 1, no difference from the genome-wide average in rate of change in locus-specific allele frequency. (c) *v* = 30, a steep cline indicating restricted gene flow across the cline *centre*, indicating reproductive isolation or ecological selection across a sharp environmental transition. Note that cline shapes for extreme cline centres are virtually indistinguishable among panels a–c. (d) Comparison of geographic (blue dashed lines) versus genomic (pink solid lines) cline shape for a steep cline (*v* = 10) with cline *centres* 0.1 to 0.9. For cline *centre* = 0.5, the two cline shapes are indistinguishable. (e) Estimated cline *v* (*gghybrid*) or 1/*width* (*hzar*, equivalent to *v*) and (f) estimated *centre*, and their 95% credible intervals (*gghybrid*) or confidence limits (*hzar*) for two independently simulated datasets for each sample size 10, 100, 1000, 10,000, all with true *centre* = 0.5, *v* = 10. Black horizontal lines indicate true *v* and *centre*. Genotypes were simulated either from the genomic (blue) or geographic (pink) cline function and fit using *gghybrid* (closed circles) or *hzar* (closed squares). For sample size = 10, the point estimate of *v* from *hzar* for one simulated dataset, and centre from hzar for the other dataset, were well outside the y axis range, and hence the estimates are not plotted.

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against hybrids, among other quantities, can be extrapolated from these parameters (e.g. Bailey et al., 2015). This two-parameter model makes several simplifying assumptions that can affect interpretation. Parental allele frequencies are assumed known and could affect cline estimates if incorrect but could additionally be estimated. Cline shape may deviate from a simple sigmoid curve for loci in linkage disequilibrium (LD) with loci under selection, forming stepped clines (Barton, 1983; Porter et al., 1997; Stankowski et al., 2019; Szymura & Barton, 1986). These models contain more parameters, including a sharp sigmoid central step formed due to LD (two parameters as before), surrounded by exponential tails of introgression whose width more closely reflects the strength of direct selection on the locus. Furthermore, gene flow across stepped clines is often likely to be asymmetric, and their estimation therefore requires at least 4 parameters in total. For symmetric stepped clines, the width of a fitted two-parameter tanh curve represents a compromise between the sharp central step caused by LD and the surrounding exponential tails. Fitting a tanh curve to an asymmetric cline may be more problematic, potentially leading to incorrect cline centre estimates and hence spurious evidence of cline displacement among loci in a hybrid zone.

Both the assumed shape of the geographic cline and any downstream extrapolations require that the hybrid zone is maintained by a balance between dispersal of parental genotypes into the hybrid zone, and selection either against hybrids or across a sharp environmental transition (Kruuk et al., 1999). There are many situations that violate these assumptions. For example, admixed populations may be isolated from their parent taxa and receive little or no gene flow (e.g. Cuevas et al., 2022; Deacon et al., 2019), the pattern of contact may be mosaic, with mixtures of parental and hybrid populations persisting in close proximity (e.g. Bridle & Butlin, 2002; Vines et al., 2003), or admixture may occur over a much larger geographic range than lifetime dispersal (e.g. Trier et al., 2014).

In 1985, Barton & Hewitt (see also Szymura & Barton, 1986) introduced the concept of genomic cline analysis to deal with patterns of contact that violate the assumptions of geographic cline analysis. In genomic cline analysis, geographic distance is replaced by a measure of whole-genome hybrid index or admixture proportion (the two terms are synonymous when there are only two source populations), which ranges from 0 (all genomic variants originate from source 0) to 1 (all from source 1). Individual loci are then tested for deviation in their change in allele frequency from the change in the genome-wide hybrid index (Figure 1a–c). The two key parameters in genomic cline analysis have previously been described as the *rate* (analogous to the inverse of cline *width*) and *location* (analogous to cline *centre*) and serve the same purpose as *width* and *centre* respectively in identifying loci of interest (Gompert & Buerkle, 2011). Genomic cline analysis therefore allows cline parameter estimation in situations where allele frequency changes in space do not represent a balance between dispersal and selection. Despite its wider applicability, one downside of using genomic cline analysis in situations that violate assumptions of geographic clines is a reduction in interpretability of cline parameters, such as through downstream estimation of the strength of selection. While spatial and other complexities are removed from the modelling framework, their impacts remain. Hence, results must be interpreted considering potential geographic structure, such as the presence of physical barriers to dispersal.

There are several existing software packages for fitting geographic or genomic clines. Although no longer available, the first publicly available cline software, implementing both genomic and geographic (including in 2 geographical dimensions) cline analysis, was *Analyse* (Barton & Baird, 1995). The R package *hzar* (Derryberry et al., 2014) fits the same one-dimensional geographic clines as *Analyse*, including stepped clines, to both genotype and phenotype data using maximum likelihood and includes useful model comparison functions. The original genomic cline function described by Szymura and Barton (1986) is now implemented in the software *bgc* (Gompert & Buerkle, 2012), which jointly estimates genome-wide hybrid index and locus-specific clines in a single model. However, in 2013, Fitzpatrick introduced an alternative 'logit-logistic' genomic cline function (Fitzpatrick, 2013), which is a simple geographic cline (tanh curve) transformed to be constrained to fall between [0,1] on the x axis, to represent genome-wide hybrid index rather than geographic distance (Figure 1a–d). The logit-logistic cline function has since been advocated by Barton as preferable to the original Szymura & Barton cline, which has some unhelpful statistical properties (Barton, 2013; Fitzpatrick, 2013; Gompert & Buerkle, 2011). The R package *HIest* (Fitzpatrick, 2013) fits the logit-logistic genomic cline to bi-allelic data but is limited to loci with fixed differences between source populations. *Introgress* (Gompert & Buerkle, 2010) is a now obsolete R package that separately estimated hybrid index and genomic clines, using its own unique cline function. The workflow of *introgress* provided inspiration for the setup of the new R package described here.

Here, I introduce the R software package *gghybrid*, which uses *Structure* (Pritchard et al., 2000) format input files for Bayesian estimation of hybrid index and Fitzpatrick's logit-logistic genomic clines. *gghybrid* fits simpler models than *bgc*, based on allele frequencies rather than genotypes, with separate functions for 1-parameter hybrid index and 2-parameter genomic cline estimation and with each individual (hybrid index) and locus (genomic clines) treated as an independent analysis. As with all existing genomic cline packages, fitting stepped clines is not yet included. Unlike Fitzpatrick's *HIest*, *gghybrid* allows parental polymorphism when estimating hybrid index and genomic clines but does not include genotype uncertainty or parental allele frequency uncertainty in parameter estimation, which are allowed in *bgc*. The functions use fast, vectorised calculations across test subjects (individuals or loci) to speed up runtime, which can be further improved by parallelising test subjects into blocks (Figure 2). Its simplicity of design allows it to scale up to whole genome data and to be used for statistical model comparison through grouping allele copies in different ways beyond the defaults (individuals and loci). It includes several methods for statistical model comparison and includes useful plotting functions. As well as calculating a Bayesian *p* value for deviation of each locus from the genome-wide average for

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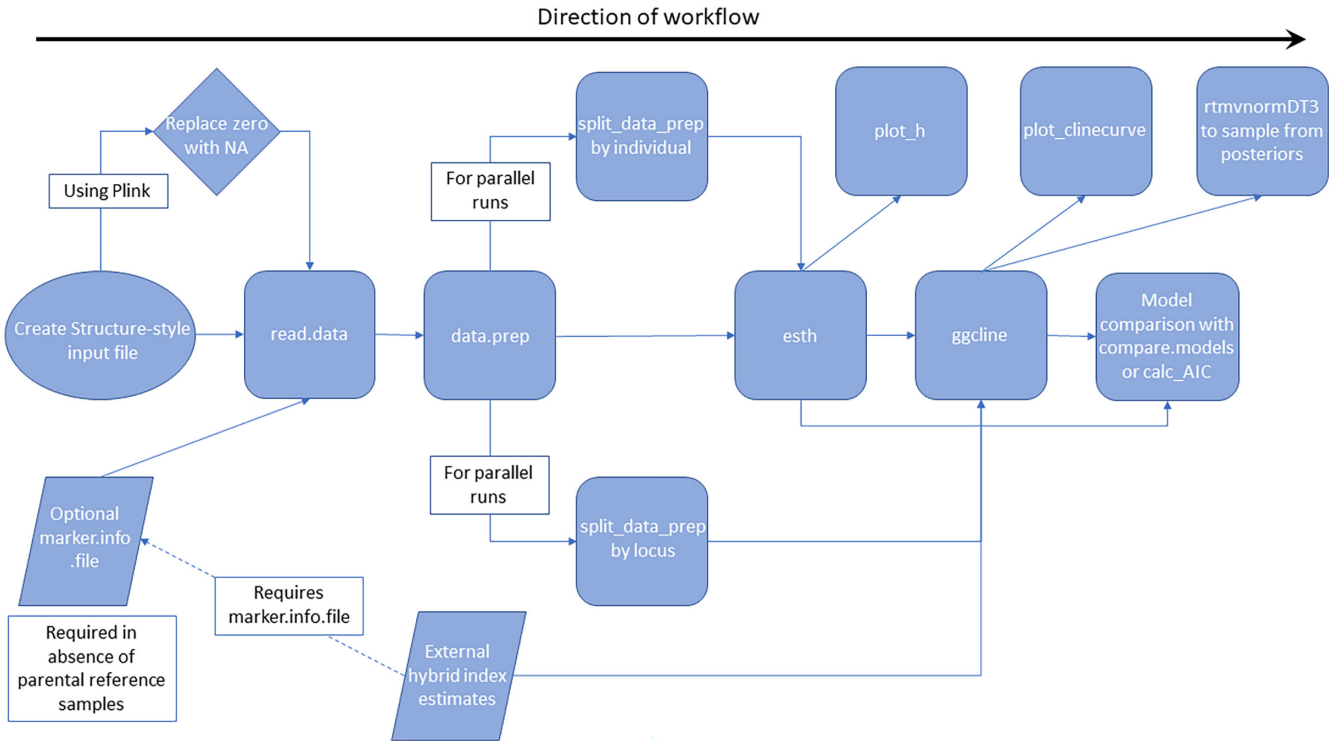


FIGURE 2 Workflow for hybrid index and genomic cline estimation in *gghybrid*. Package functions are in round-cornered boxes. The primary workflow is depicted in the centre, from left to right. The 'esth' function estimates hybrid index, and the 'ggcline' function estimates genomic clines.

each parameter, it includes functions to estimate two information criteria, AIC (Akaike, 1998) and waic (Watanabe & Opper, 2010) for model selection. Admixture proportions estimated in other software (Schubert et al., 2020) can also be used in genomic cline analysis when accompanied by allele frequency estimates for each locus in the two source populations.

2 | METHODS AND RESULTS

2.1 | Overview and installation of *gghybrid*

gghybrid version 2.0.0 can be installed directly from Github (ribailey/gghybrid: R package for analysis of hybrids and hybrid zones (github.com)) using one line of code in R: devtools::install_github("ribailey/gghybrid"). The package and previous versions are stored permanently at Zenodo, DOI: 10.5281/zenodo.3676498. Documentation includes the typical R package help files, and the Github and Zenodo uploads also include an extensive example code file, code for all the data simulations and analyses presented here, and a formatted example dataset of publicly available sparrow transcriptomic SNPs, used by the author in an earlier study (Hermansen, Haas, Trier, et al., 2014) and downloaded from Dryad data repository (Hermansen, Haas, Bailey, et al., 2014; https://doi.org/10.5061/dryad.v6f4d).

gghybrid includes functions to (see Figure 2): (i) read in data in the form of *Structure* files or similar; (ii) prepare data for analysis, identify parental reference samples, calculate parental allele frequencies, and carry out locus filtering based on parental reference sample quantities including parental sample sizes, allele frequency difference between parental reference populations, and parental minor allele frequency; (iii) split the analysis object by individuals or loci if desired, in preparation for parallel or serial runs of either hybrid index or genomic cline estimation; (iv) estimate hybrid index; (v) estimate genomic clines; (vi) plot hybrid index and genomic cline results; (vii) sample from the posterior distributions of fitted genomic clines; (viii) calculate the Akaike Information Criterion (AIC) for hybrid index genomic cline model fits; and (ix) calculate the widely applicable information criterion (waic) and its difference between pairs of models.

To increase memory efficiency when estimating hybrid index or genomic clines across many test subjects, *gghybrid* uses recursive updating of posterior parameter means, variances and covariance (Pebay, 2008), mcmc (Monte Carlo Markov Chain; Brooks et al., 2011) acceptance rates (Ovaskainen et al., 2008), and posterior likelihoods, storing only posterior means and (co) variances each iteration. For ease of use, *gghybrid* employs an adaptive burn-in phase to optimise mcmc sampling (Gelman et al., 2004; Ovaskainen et al., 2008), therefore minimising the required number of pre- and post-burn-in iterations, rarely requiring changes to default settings, and normally requiring only a single run per analysis. The option to make real-time mcmc point plots of parameter values for multiple test subjects is included for diagnostics, and convergence of multiple runs can be tested post hoc through sampling from posterior distributions. All output tables

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in *gghybrid* are data.tables from the *data.table* package (Dowle & Srinivasan, 2021), allowing rapid and memory-efficient post-analysis manipulations of output objects for those familiar with this package. However, data.tables are also data.frames and can be treated syntactically the same, and they are also compatible with the *tidyverse* pipeline (Wickham et al., 2019).

2.1.1 | Hybrid index and genomic cline formulae

The estimated genome-wide hybrid index h of a test subject represents the proportion of its genome originating from declared parental reference 'S1' as opposed to 'S0'. As previously described (Buerkle, 2005), the probability of an individual allele copy having originated from source S1 therefore depends on the frequency of the allele in each source population and the hybrid index of the test subject, and is given by $p_{ij} = hS1_{ij} + (1 - h)S0_{ij}$, where $S0_{ij}$ and $S1_{ij}$ represent the allele frequency for the jth allele of the ith locus in each parental reference sample. Each allele copy k is treated as independent by *gghybrid*, and the log likelihood across the n allele copies for each test subject is

$$L(h|\ allele\ identity) = \sum_{k=1}^{n} \left(ln\left(hS1_{ij} + (1-h)S0_{ij}\right)\right) + \ln(\pi(h))$$

where $\ln(\pi(h))$ refers to the log likelihood of the beta-distributed prior for h, with user-specified beta shape parameter values (default is Jeffrey's prior, which is symmetric, U-shaped with mean at 0.5 and *shape 1* = *shape 2* = 0.5). This results in a beta-distributed posterior for h.

The logit-logistic genomic cline is a transformation of the standard tanh geographic cline, such that the predictor of locus-specific allele frequency (genome-wide hybrid index) is constrained to [0,1]. The locus-specific frequency A of the allele $j1$ with higher frequency in source S1 than S0 is $A_{j1} = S0_{j1} + (h^v/(h^v + (1 - h)^v u^e))(S1_{j1} - S0_{j1})$, where v is cline steepness (rate) and u is cline location, both relative to the genome-wide hybrid index ($v = 1$ and $u = 0$ indicate no deviation from the genome-wide average). Parameter v is equivalent to $1/width$ of a cline back transformed onto the unconstrained geographic cline scale, and therefore represents the maximum slope. Parameter u is analogous to geographic cline centre, but in this formulation its range of values scale to v (Figure S1) and hence I estimate and present the more interpretable cline *centre* (genome-wide hybrid index at which the locus-specific allele frequency is halfway between those of S1 and S0), where $\text{logit}(centre) = u/v$.

gghybrid jointly samples and estimates $\ln(v)$ and $\text{logit}(centre)$, so that the joint posterior on this latent scale is multivariate normal. The joint log likelihood for the two parameters v and u (θ) is

$$L(\theta|\ allele\ identity) = \sum_{k=1}^{n} \left(\ln\left(\left(\frac{h^v}{h^v + (1-h)^v u^e}\right)S1_{ij} + \left(1 - \left(\frac{h^v}{h^v + (1-h)^v u^e}\right)\right)S0_{ij}\right)\right) + \ln(\pi(\theta))$$

Where $\ln(\pi(\theta))$ is a multivariate normal joint prior on $\ln(v)$ and $\text{logit}(centre)$. *gghybrid* estimates the multivariate normal posterior latent parameter means, variances and covariance.

2.1.2 | Interpretation of parameter values

The [0,1] constraint on the x axis creates a cline in locus-specific allele frequency against genome-wide hybrid index that can take a variety of shapes (Figure 1a–c). Only when $v > 1$ does the curve approximate the typical logistic shape of a geographic cline (Figure 1c,d), indicating restricted gene flow either side of the cline *centre*. The cline shape differs from the original Szymura & Barton cline and the parameters are not directly comparable (Fitzpatrick, 2013). If $v = 1$ (Figure 1b), there is no difference in rate of change between locus-specific allele frequency and genome-wide hybrid index, while $v < 1$ (Figure 1a) indicates both alleles have introgressed excessively into the foreign genomic background. Cline *centre* = 0.5 indicates no directional bias in favour of introgression of either allele. Cline *centre* > 0.5 indicates a bias favouring introgression of the S0 allele into the S1 genomic background, and vice versa for *centre* < 0.5 (Figure 1a–d).

Extreme cline *centre* is a measure of biased introgression, but does not necessarily indicate that an allele has swept to fixation (e.g. in Figure 1a,b individuals with intermediate hybrid index remain polymorphic for a wide range of *centre* values), as this scenario also depends on v and on parental allele frequencies. Nevertheless, for $v \geq 1$ (or v fixed at 1) it clearly indicates directional bias. Additionally, the user can estimate locus-specific hybrid index (LSHI, see Section 2.2.3).

2.2 | *gghybrid* pipeline description

The *gghybrid* workflow is summarised in Figure 2.

2.2.1 | Reading in data—the *read.data* function

The input file format includes all existing *Structure* formats and is further generalised to all tables with a full, partial or no header row and with allele copies per locus organised into either multiple adjacent columns or multiple adjacent rows. As with *Structure* files, non-locus columns (any number are allowed) must be to the left of locus columns, and the minimum requirement is a column containing a unique reference for each hybrid index test subject. Typically, each test subject is an individual (but could be another level of organisation) and the default column name is 'INDLABEL', which is flexible. By default, a second column, 'POPID', is expected, which identifies the population of origin and is useful for declaring parental reference populations downstream. Existing *Structure* files and their accompanying MAINPARAMS files can mostly be used without changes, although there are limitations to the spelling of entry names if a separate MAINPARAMS file is uploaded. The same MAINPARAMS information can also be input directly in *read.data*. The "–*recode*

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structure" function in *Plink* (Chang et al., 2015) can be used to create the *gghybrid* input file, for example for large whole-genome datasets. However, it produces a *Structure* format file with missing data coded as zero. At the time of writing, the underlying *data.table* package does not allow zeroes to be declared as missing data in *read.data*, and hence these should be recoded as 'NA' before using *gghybrid*'s *read.data* function. This limitation has been actioned and may be fixed in later versions of *data.table*.

All ploidies are allowed. For datasets with varying ploidy among markers, the input object should contain the same number of entries for all markers, with non-existent allele copies coded either as missing data or as duplicates of true entries (such as coding mitochondrial loci as diploid homozygotes when input alongside diploid nuclear loci). All allele copies coded as missing data, whether due to missingness or ploidy variation, are removed from the dataset prior to analysis. Hence, in the case of ploidy variation, loci with lower ploidy will have lower statistical power in both hybrid index and genomic cline estimation.

A separate 'marker info' file can also be uploaded at this stage, containing one row per marker with marker-specific information that may be useful downstream, such as for hypothesis testing. In the absence of parental reference samples, this file is required and must include parental allele frequencies. Examples of potentially useful marker information are functional annotations or identifiers of windows along chromosomes.

The primary output is a table with a full header row, one column per marker (plus any non-marker columns) and one row per test subject per allele copy. The marker info file is included as a separate table in the output object, ready to be joined to the analysis table at the next stage if desired.

2.2.2 | Preparation of data for analysis—the *data.prep* and *split_data_prep* functions

The primary purpose of the *data.prep* function is to prepare a long-form table, with one row per non-missing allele copy in the dataset, containing all locus-specific information required for statistical analysis. Primary inputs for *data.prep* come from the *read.data* output object. If parental reference ('Source', S0 and S1) samples are present, they should be identified here, and all filtering based on parental data is carried out by this function. Parental allele frequencies are either taken from the marker info file previously uploaded, or calculated internally using the supplied and named parental reference samples as the mode of a resulting posterior beta distribution with a uniform prior. Parental reference sample IDs are by default taken from the POPID column, but any alternative column (such as INDLABEL) can be used, by declaring it as the 'POPID.name'.

As well as the analysis table, the user can optionally output a table with one row per locus and a larger array of locus-specific variables in columns, such as the parameters of the beta-distributed estimates of parental allele frequencies. Furthermore, *data.prep* can output a genotype table with genotypes scored from zero to ploidy, and scores indicating the number of copies of the allele with relatively higher frequency in the declared source 'S1' than in 'S0'. This may be useful for using the covariance formula (Rogers & Huff, 2009) to calculate admixture-induced linkage disequilibria (e.g. Grossen et al., 2016).

The recommended locus filtering option is 'AF.CIoverlap = FALSE', which removes all loci for which there is overlap between S0 and S1 in their Bayesian posterior 95% credible intervals of allele frequency. This ensures high confidence that allele frequency differences between parental samples are real and in the estimated direction. Including large numbers of loci with low-confidence allele frequency differences may add spurious noise to hybrid index estimates, and these loci are also of little use in genomic cline analysis. Note that loci with exactly zero allele frequency difference between parental reference samples are removed even without any filtering.

At this stage, the *split_data_prep* function can optionally be used to divide up the analysis table in preparation for parallel (or sequential) runs of either hybrid index or genomic cline estimation. The column by which to split the data can be chosen (for example 'INDLABEL' for hybrid index estimation, or 'locus' for genomic cline estimation), alongside the number of test subjects desired within each sub-file. The resulting objects are saved to the working directory as csv files and removed from the R workspace. Each file can then be loaded and used as the *data.prep* data analysis object for downstream functions. The chosen test subject is included in the sub-file names, making them easy to identify and list.

2.2.3 | Hybrid index estimation—the *esth* and *plot_h* functions

Inputs for *esth* are primarily output objects from the previous functions. Both the prior and posterior for hybrid index are beta-distributed, with default Jeffrey's prior (*shape1* = *shape2* = 0.5). Both the posterior mode ('h_posterior_mode') and mean ('beta_mean') are presented in the output alongside the posterior variance, 95% credible intervals, and beta shape parameters. The mode is then used by default as the hybrid index for genomic cline estimation, with the option to switch to the mean. For the default prior, both the mean and mode can be treated as the best posterior estimate, and the mean has the potentially useful property of always falling between 0 and 1 (the mode can be exactly 0 or 1). If the prior is excluded from the analysis by setting prior beta *shape1* = *shape2* = 0, the posterior mean is equivalent to the maximum likelihood. With a uniform prior (*shape1* = *shape2* = 1), the posterior mode is equivalent to the maximum likelihood. Accurate, convergent posterior estimation can be achieved with as little as burn-in = 1000, nitt (total number of iterations including burn-in) = 3000. Multiple runs can be carried out if desired, and instructions for testing convergence using the Gelman diagnostic (Gelman & Rubin, 1992) are included in the example code. Note that the Bayesian posteriors are beta-distributed, which violates the assumption of normality in the Gelman diagnostic. Extreme high or low hybrid index estimates may therefore

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artifactually produce poor Gelman diagnostic values, even when logit-transformed to improve normality, possibly due to skewness. The user can optionally make real-time mcmc point plots of posterior values for one or more test subjects, to check the efficiency of mcmc.

The default test subject is 'INDLABEL', the individual reference, but any character or factor column in the data analysis table can be chosen, such as a population reference, windows along a chromosome, locus, and so forth. Hence, it is possible for example to estimate a locus-specific hybrid index (LSHI) across all samples simply by setting 'test.subject = "locus"'. Note that by default, parental reference S0 and S1 as well as TEST individuals are included in the analysis, so for example to estimate LSHI across 'TEST' individuals only, set 'include.Source = FALSE'.

Hybrid index estimates made at different resolutions can be compared using AIC and waic. For example, statistical comparison of independent individual-level estimates with a single population-level estimate would reveal the extent of evidence for intra-population variation in hybrid index. This may be expected if a hybrid population has not yet reached equilibrium due to recency of contact, or there is ongoing gene flow from differentiated populations such as in the centre of a hybrid zone (Barton & Gale, 1993). If model comparison using waic is desired, set the *esth* option 'return.likmeans = TRUE' to produce an output table of likelihood means and variances per datapoint. There are downstream functions for calculating each information criterion. In the case of waic, the *compare.models* function is designed to compare two models. An entry for the *esth* option 'test.subject.compare' should be declared in advance of using *compare.models*. 'test.subject.compare' within *esth* indicates the test subject of a lower-resolution model with which the current model will be compared. 'test.subject.compare' therefore only needs to be declared when currently estimating the higher resolution test subject (e.g. individual reference) and can be ignored when running the lower-resolution test subject (e.g. population reference).

The *plot_h* function in *gghybrid*, which uses base R graphics, includes several useful built-in options that can add value to the plots (Figure 3).

2.2.4 | Genomic cline estimation—the *ggcline*, *rtmvnormDT3* and *plot_clinecurve* functions

The primary inputs for the main genomic cline function, *ggcline*, are outputs from *read.data*, *data.prep* and *esth*. As with *esth*, analysis can be run without changing defaults, however by default *ggcline* does not include parental reference (S0 and S1) individuals in genomic cline estimation, only those declared as 'TEST' individuals. It may often be useful to include S0 and S1 individuals, and this can be done by setting 'include.Source = TRUE'. As with *esth*, if model comparison using waic is desired, set 'return.likmeans = TRUE'. Testing has indicated that burn-in = 2000, nitt = 5000 are sufficient for accurate and convergent parameter estimation. The user can optionally make real-time mcmc point plots of *v* and logit(*centre*) for one or more test subjects, to check mcmc efficiency. *ggcline* outputs the parameter estimates and their 95% credible intervals for each test subject on the original scale, plus the posterior parameter means, variances and covariance on the latent scale, and a Bayesian *p* value for the deviation of each parameter from the genome-wide expectation.

Default (random) starting values and priors work well, but both can be set by the user. User-defined starting values are on the original

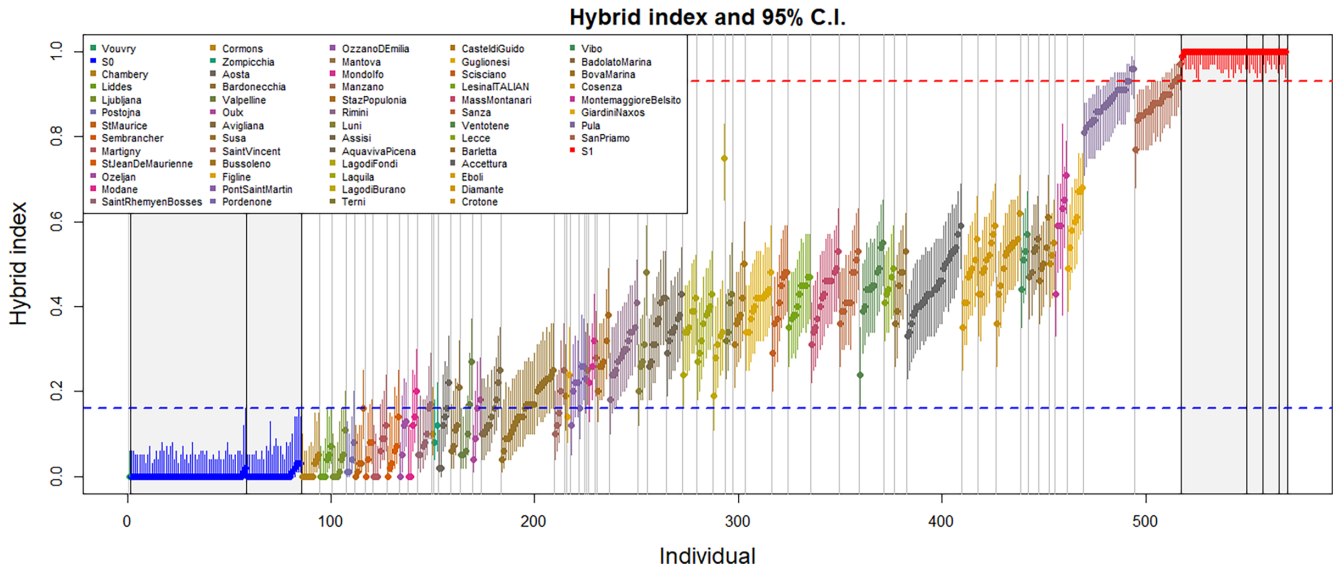


FIGURE 3 Example hybrid index plot using *plot_h*, using Italian sparrow transcriptomic SNP data from Hermansen, Haas, Trier, et al. (2014). Points are hybrid index posterior mode, and error bars are 95% credible intervals. Points are ordered by population mean then individual estimate. Grey shading indicates parental reference samples. Vertical lines are population boundaries. Horizontal dashed lines show the inner credible interval limits for the two parental reference samples.

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scale (*v* ranges 0-infinity with null value 1, while $0 < centre < 1$, with null value 0.5). The priors are specified on the latent parameters ln(*v*) and logit(*centre*), with null value of zero in both cases.

As with *esth*, model comparison can be set up in *ggcline*. The default test subject (in this case 'locus') can be changed, and different test subjects on the same data can be compared using the two information criteria. Each parameter can individually be pooled across all test subjects, hence estimating a single value for the pooled parameter. When pooling, it is recommended to use at least *burn-in = 5000*, *nitt = 10,000* due to reduced mcmc efficiency. If pooling across subsets of test subjects is desired, each subset must be run separately. Either or both parameters can also be fixed to the null or any other value by declaring either a single value to apply across all test subjects, or a unique value for each.

Random samples can be taken from the joint posterior distribution of the fitted cline parameters for all or any subset of test subjects, using the function *rtmvnormDT3*. This function can be used more generally to sample from a set of bivariate normal distributions (including with truncation) organised in a similarly formatted data. table as the *ggcline* results table. As described in the example code file, the Gelman diagnostic can be calculated on multiple runs to test for convergence, using these post-hoc random posterior samples. However, this is rarely likely to be necessary.

The *plot_clinecurve* function uses base R graphics to plot fitted clines for one or more loci and has several useful options. Though not built into the function, data points can be added to the plot, as can multiple cline curves representing uncertainty, from random sampling of the posterior using *rtmvnormDT3*, as described in the example code file (Figure 4).

2.2.5 | Model comparison—the *compare.models* and *calc_AIC* functions

The two model comparison functions are designed to run multiple model comparisons simultaneously. For both hybrid index and genomic cline results, a primary objective is to compare analysis outcomes for data split into categories of differing resolution. For example, to ask whether hybrid index estimated at the individual level is more optimal than a single estimate across all individuals within a population, for multiple populations simultaneously. In this case, model comparisons are made at the lower resolution (population). Further types of comparison are allowed for genomic cline results, for example comparison of models at the same resolution but with differing combination of parameters estimated versus fixed in each model. Finally, one or both parameters, *v* or *centre*, can be pooled to obtain a single estimate across the whole dataset, and a statistical comparison made between models with different pooling at the level of the whole dataset.

The function *compare.models* calculates waic and its difference for a pair of models across all test subjects for the lower resolution model. To use *compare.models*, the option 'return.likmeans = TRUE' must have been set when calling the *esth* or *ggcline* functions. The value for waic is scaled to be directly comparable to AIC, where $AIC = -2LL + 2p$ and $waic = -2lppd + 2p$, where *LL* is the log likelihood of the best posterior parameter estimate(s) given the data, and *lppd* is log pointwise predictive density (Vehtari et al., 2017). While the *compare.models* function includes calculation of the effective number of parameters (which is typically used in the waic calculation), and presentation of this value and the resulting waic in columns including 'effpar' in the name, it is recommended to instead use waic calculated with the actual number of parameters (columns named '…npar…'). This is because testing indicates that the effective number of parameters is systematically overestimated. Invalid estimates of the effective number of parameters may be a result of posterior likelihood variances for some data points being too large (Vehtari et al., 2017).

The *calc_AIC* function calculates AIC for a single *esth* or *ggcline* output object, two or more of which can then be joined and compared manually, following instructions in the example code file.

2.3 | Statistical properties of Bayesian hybrid index and genomic cline estimation

2.3.1 | Comparison of *gghybrid*'s hybrid index estimation with *introgress* and *bgc*

In studies of a crow hybrid zone, Knief et al. (2019) found a very high correlation ($r \geq 0.99$) between hybrid index estimates from *gghybrid* and the *introgress* package, which uses the same likelihood formula. Similarly, I found $r > 0.999$ between the two packages using the Italian sparrow dataset from Hermansen, Haas, Trier, et al. (2014). I also found $r > 0.99$ between *gghybrid* and previous hybrid index estimates using *bgc* for Italian sparrows (Hermansen, Haas, Trier, et al., 2014).

2.3.2 | Comparison of *gghybrid*'s genomic cline analysis with *hzar* and *bgc*

To compare *gghybrid* with *bgc*, I repeated the Italian sparrow genomic cline analysis previously run in *bgc* by Hermansen, Haas, Trier, et al. (2014). The *bgc* and *gghybrid* genomic cline functional forms differ and hence parameters may not be directly comparable. However, their identification of significant clines can be compared using overlap of posterior 95% credible intervals with the null value as a significance test for each parameter and software. All loci with significantly steep clines in *bgc* were also significant in *gghybrid*, while *gghybrid* identified more significantly steep clines overall (Figure S1). *bgc* estimated some clines to be shallow (*beta* parameter below zero) and non-significant for several loci that were significantly steep in *gghybrid*. For deviation of cline *centre* (*alpha* parameter in *bgc*) from the null there was also considerable overlap between *gghybrid* and *bgc*, although a few loci were significant in only one of the two packages.

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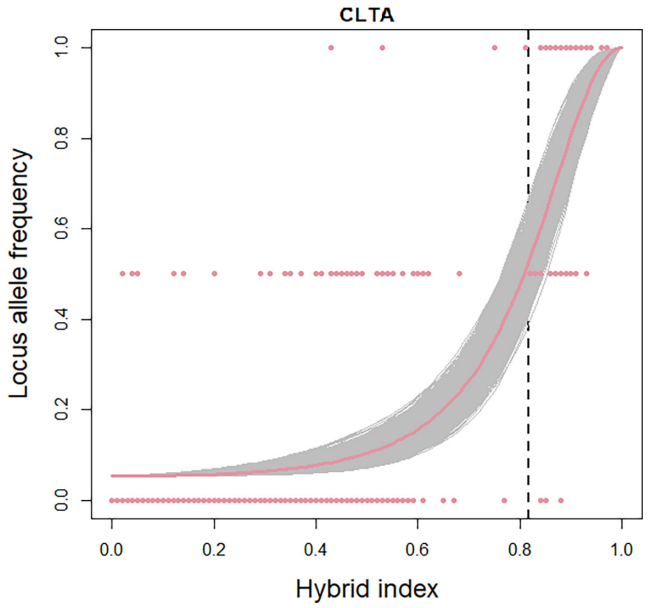


FIGURE 4 Example cline plot using *plot_clinecurve*, for the CLTA locus in the Italian sparrow dataset from Hermansen, Haas, Trier, et al. (2014). The function can plot cline curves for one or several loci, while the data points (diploid genotypes, pink points) and uncertainty (grey lines from 1000 random posterior samples of cline parameters) were added manually, following instructions in the example code file. The vertical dashed line is the best-fitting cline centre.

For cline *centre* = 0.5 and a steep cline falling fully within the range [0,1], geographic and logit-logistic genomic cline curves are indistinguishable (Figure 1d), allowing direct comparison between the R packages *hzar* and *gghybrid*. I simulated eight independent sets of haploid individuals for one locus with *centre* = 0.5 and *width* = 0.1 (therefore $v = 10$, as $v = 1/width$), four using the geographic cline function from *hzar* and four using *gghybrid*'s genomic cline function, one each for haploid sample sizes 10, 100, 1000 and 10,000. I then used both *hzar* and *gghybrid* to estimate cline parameters. The two packages produced comparable cline parameter estimates for sample sizes 100 and above, while *hzar* produced inaccurate estimates and either very wide or very narrow confidence limits for sample size 10 (Figure 1e,f).

2.3.3 | Statistical properties of Bayesian logit-logistic genomic cline analysis

To test the rate of statistical false positives among loci with true clines that do not deviate from the genome-wide average (*centre* = 0.5, $v = 1$) by any mechanism such as selection or drift, I simulated genotypes for 1000 diploid loci fixed in parental populations, all with null true cline parameters, across 1000 individuals with evenly spaced hybrid indices from 0 to 1. For cline *centre*, 4.6% of loci were false positives according to the Bayesian *p* value (threshold $p \leq .05$), while for cline *v*, 6.3% were false positives. For both AIC and waic, 5.1% of loci were false positives for *centre* and 4.3% for cline *v*, using a threshold of ΔAIC/Δwaic ≥2 for the one-parameter model (either *centre* or *v* estimated) being a better fit than the null model (both parameters fixed at null values).

To identify loci of interest, the user has the choice between Bayesian *p* values and parameter posterior distributions from the full model alone, versus ΔAIC/Δwaic from model comparisons. To compare these methods, I simulated 980 diploid loci across 1000 individuals with evenly spaced hybrid indices from 0 to 1, for each of (i) *centre* = 0.5 and ln(*v*) evenly spaced between −4 and 4 (20 loci for each of 49 values, including the null ln(*v*) = 0), and (ii) $v = 1$ and *centre* evenly spaced from 0.000001 to 0.999999 (20 loci for each of 49 values, including the null *centre* = 0.5). Using a threshold of $p \leq .05$, or ΔAIC/Δwaic ≥2, all methods performed well in identifying deviations from null (Figure 5). However, the change in *p* value was not fully consistent with the magnitude of parameter deviation from the null value. Hence, *p* value alone should not be used to rank loci beyond whether they meet a (not too stringent) threshold criterion. Further ranking should therefore be done using the parameter estimates when only running the full model. Loci can, however, be ranked effectively by ΔAIC/Δwaic, and use of waic is strongly recommended.

To test the efficiency of AIC and waic when both parameters may deviate from null, and to test prediction accuracy with the full model, I simulated 2401 diploid loci on an evenly spaced grid of *centre* (0.000001 to 0.999999; 49 values including *centre* = 0.5) and ln(*v*) (−4 to 4; 49 values including $v = 1$), again with 1000 evenly spaced hybrid indices, and ran all four cline models (full model, each parameter fixed, both parameters fixed). Cline *centre* estimates (Figure 6a) were most accurate for steep clines (ln(*v*) > 2) but *centre* could not be accurately estimated with very shallow clines (ln(*v*) < −2). However, the true value for *centre* remained consistently within the posterior credible intervals across all values of *v* (Figure 6c), indicating wide credible intervals for *centre* when ln(*v*) < −2. Mean deviation of the best posterior estimate of *v* from the true value increased with increasing true *v* (Figure 6b). However, the true value fell outside the posterior credible intervals more often for small values of *v* (Figure 6d). For the most extreme cline *centre* values and $v \geq 1$, different true *v* could not be distinguished, probably due to extreme distortion of cline shape (see Figure 1a-c).

All of AIC (Figure S2a,b), waic (Figure S2c,d) and *p* value (Figure S2e) effectively identified deviations in cline *v* from null. waic was most effective and *p* value least effective at identifying deviations from null cline *centre* for shallow cline *v*, while all methods were effective for steep clines. However, results indicated there may be false positives for cline *centre* using waic for very shallow clines (ln(*v*) < −2).

2.3.4 | Impacts of an uneven distribution of hybrid index values on genomic cline analysis

Genomic cline analysis is most effective when the test population includes a high resolution and even distribution of hybrid index values. To test the effect of uneven hybrid index distribution, I simulated

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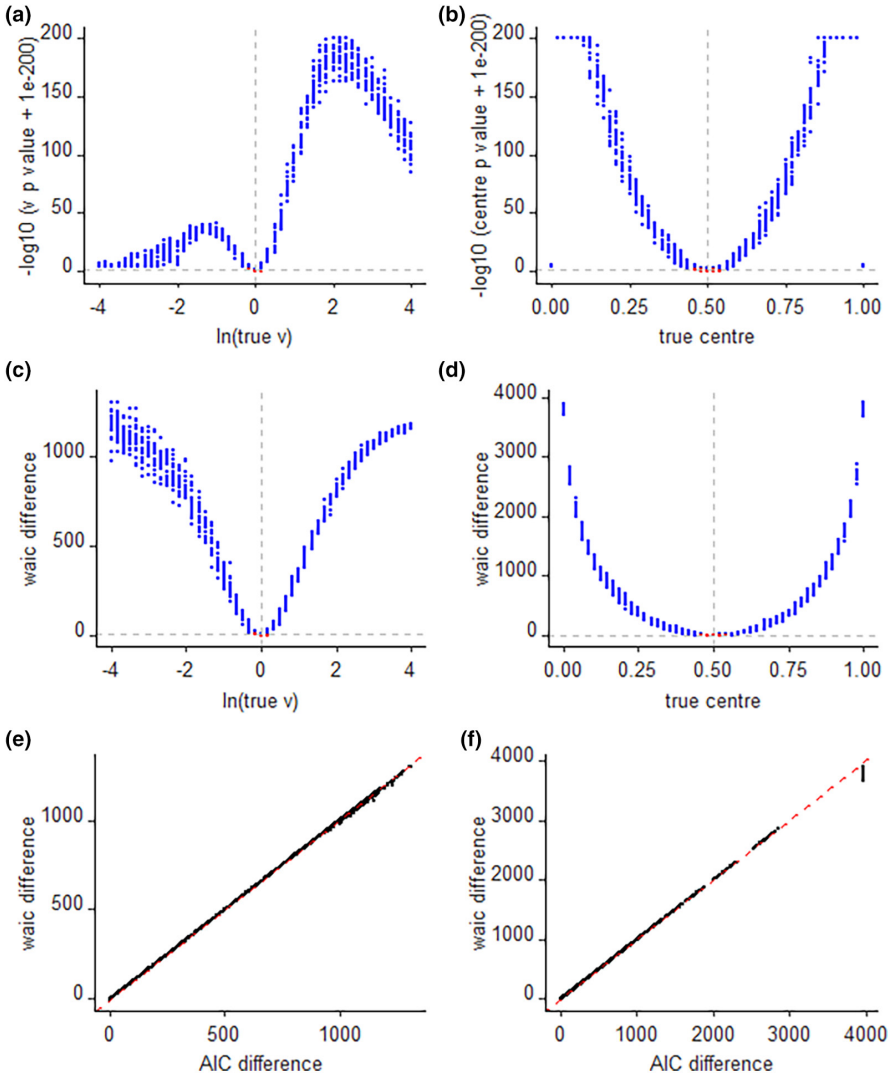


FIGURE 5 Genomic cline statistical properties with one non-null parameter. (a) −log10(Bayesian *p* value + constant) for 980 simulated diploid loci with *centre* = 0.5 and ln(*v*) ranging −4 to 4, with genotypes sampled from 1000 individuals with evenly distributed hybrid indices. Red points did not pass threshold criteria ($p < .05$). (b) As for (a), but with $v = 1$ and *centre* evenly spaced from 0.000001 to 0.999999. Red points did not pass threshold criteria ($p < .05$). (c) Difference in widely applicable information criterion (Δwaic) between a null model with both parameters fixed, and a one-parameter model for the simulated dataset with varying *v*. Red points did not pass threshold criteria (Δwaic>2). (d) As (c) but for the dataset with varying *centre*. Red points did not pass threshold criteria (ΔAIC >2). (e) Comparison of Δwaic with ΔAIC for the dataset with varying *v*. Red dashed line indicates proportionality. (f) Comparison of Δwaic with ΔAIC for the dataset with varying *centre*. Red dashed line indicates proportionality.

and estimated clines for 1000 diploid loci, once with $v = 1$ (no deviation in cline steepness from the genome-wide average) and once with $v = 30$ (very steep cline) across all loci, in each case with evenly spaced true locus cline *centres* from 0.0001 to 0.9999, using 1000 simulated individuals with an uneven distribution of hybrid indices. For null cline steepness ($v = 1$) clustering of hybrid indices had little impact on prediction accuracy (Figure 7a,b), but for a steep cline ($v = 30$) both parameters were affected (Figure 7c,d). Estimated cline *centres* were highly clustered and *v* estimates were less accurate and with higher uncertainty for loci whose steep clines were centred in gaps in hybrid index. This led to an uneven distribution of loci with respect to true and estimated cline *centre* passing a threshold for *v* of $p \leq .05$ and $v > 20$ (Figure S3).

2.3.5 | Runtime

To test runtime with changing numbers of loci, I simulated null genomic clines for 1, 10, 100, 1000 and 10,000 haploid loci across 100 individuals, and ran *ggcline* on a HP ZBook Create G7 Notebook PC, with 32Gb RAM and an Intel(R) Core(TM) i9-10885H CPU @ 2.40GHz processor. Using R's *system.time* function, the increase in elapsed time per locus from the baseline for a single locus dropped with increasing number of loci (nitt = 5000, burn-in = 2000; baseline: 134.5 s; 10 loci: 2.5 extra seconds per extra locus [ESEL]; 100 loci: 1.5 ESEL; 1000 loci: 1.1 ESEL; 10,000 loci: 0.9 ESEL). The total runtime for 10,000 loci was 2 h 36 min.

3 | DISCUSSION

Genomic cline analysis is applicable in a wide variety of circumstances and represents an ideal inference method to identify candidate loci under different forms of selection in admixed populations. The R software package *gghybrid* uses simple Bayesian models to estimate Buerkle's hybrid index and Fitzpatrick's genomic clines and includes functions for plotting and hypothesis testing. It is easy to instal and run, allows widely used (*Structure*) input file formats, has good statistical properties, is flexible enough to be employed on both small and large datasets, and through parallelisation can scale up to millions of loci across whole genomes (Friedrich et al., n.d.). Runtime per test subject improves as more

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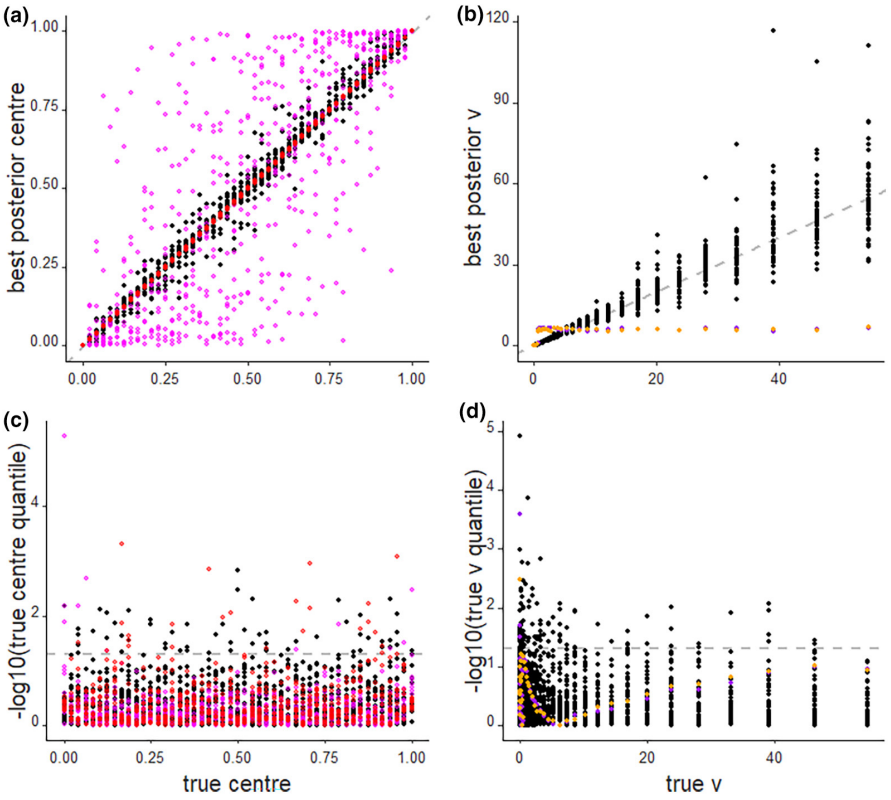


FIGURE 6 Genomic cline prediction accuracy when both parameters vary in simulations. (a) Best posterior *centre*. Red points, ln(*v*) > 2; black points, ln(*v*) ≤ 2 and ≥ −2; pink points, ln(*v*) < −2. Grey dashed line indicates perfect fit. (b) Best posterior *v*. Orange and purple points indicate the two most extreme cline *centres* (0.000001 and 0.999999). Grey dashed line indicates perfect fit. (c) −log10(quantile) of true *centre* with respect to the posterior distribution. Points above the grey dashed line fell outside the 95% posterior credible intervals. Colours are as in (a). (d) −log10(quantile) of true *v* with respect to the posterior distribution. Points above the grey dashed line fell outside the 95% posterior credible intervals. Colours are as in (b).

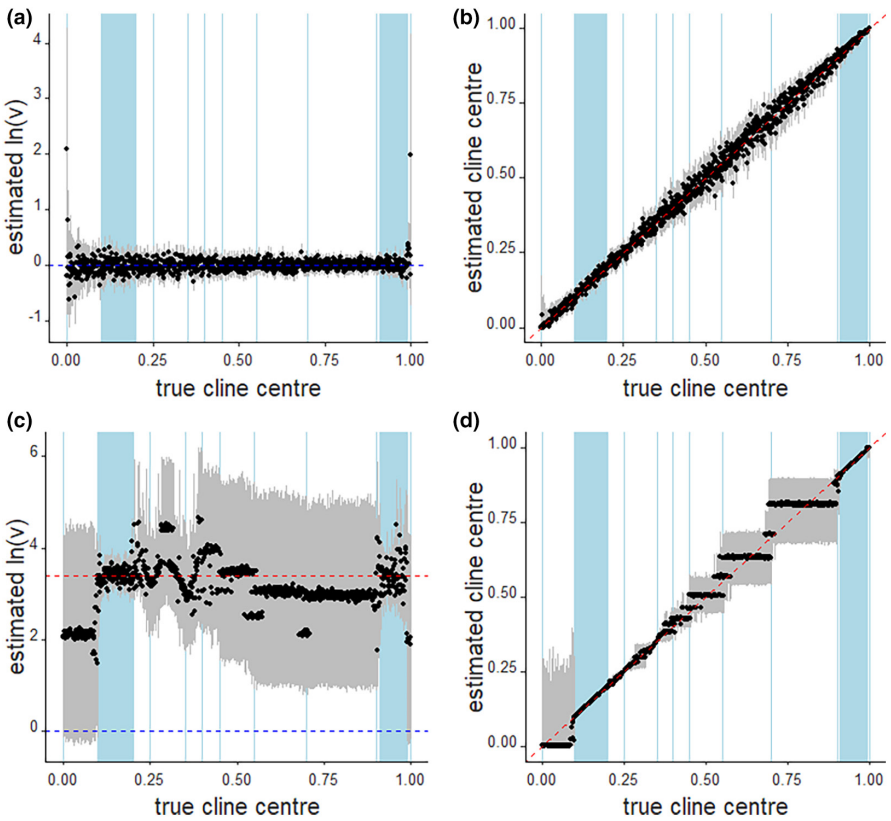


FIGURE 7 Effect of an uneven distribution of hybrid indices on posterior prediction accuracy (black points) and 95% credible intervals (grey vertical lines) for 1000 simulated diploid loci. Pale blue vertical bars indicate hybrid index values present in the simulated dataset. (a) Change in estimated ln(*v*) with changing true cline *centre*, when *v* = 1. Blue dashed horizontal line indicates true (null) ln(*v*). (b) Change in estimated cline *centre* with changing true *centre*, when *v* = 1. Red dashed line indicates true *centre*. (c) Change in estimated ln(*v*) with changing true cline *centre*, when *v* = 30. Blue dashed horizontal line indicates null ln(*v*); red dashed line indicates true ln(*v*). (d) Change in estimated cline *centre* with changing true *centre*, when *v* = 30. Red dashed line indicates true *centre*.

are added due to fast, vectorised calculations across all test subjects simultaneously and can be further improved by parallelisation. The package can therefore be readily incorporated into many genetic studies of pairs of currently or recently admixing populations or taxa.

While *gghybrid* fits a simple 1- or 2-parameter model separately to each individual (hybrid index) or locus (genomic clines), the currently most-used package for genomic cline estimation, *bgc*, fits a complex unified hierarchical model that simultaneously estimates genome-wide hybrid index and individual-locus genomic clines

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across the whole dataset (Gompert & Buerkle, 2011). This hierarchical framework leads to 'shrinkage' towards null values for poorly supported deviations from the genome-wide average at individual loci. Combined with a sum-to-zero constraint, this may lead to differences in cline parameter estimates between *gghybrid* and *bgc* beyond those due to the different cline functions. Furthermore, unlike *bgc*, *gghybrid* does not currently allow incorporation of genotype uncertainty in cline analysis.

For simplicity, uncertainties in parental allele frequencies and (for cline analysis) genome-wide hybrid index are also excluded from parameter estimation. Although not thoroughly examined, joint estimation of parental allele frequencies, hybrid index and genomic cline parameters may lead to overfitting and multiple solutions. This is because in genomic cline analysis, hybrid index represents the predictor variable in a nonlinear regression, and uncertainty in the predictor leads to a model II regression and potential for overfitting (Sokal & Rohlf, 1995). However, allowing uncertainty in parental allele frequencies may be less problematic, and will be included in future developments.

The simplicity of *gghybrid*, including separate estimation of hybrid index and genomic clines, has several advantages. It allows use of different sets of loci for the two functions, for example LD-pruned locus sets for hybrid index estimation, followed by genomic cline estimation on all loci to maximise genome coverage. Parallelisation is improved, as the analysis file can be split for example by individual for hybrid index estimation and by locus for genomic cline estimation. Separation of the two analyses allows hypothesis-testing to be carried out independently on hybrid index and genomic clines, and the data analysis table format allows allele copies to be grouped in any way for model fitting and comparison. Finally, separation of the two functions allows use of user-defined hybrid indices and parental allele frequencies in genomic cline estimation. Admixture proportions can therefore be estimated in other software.

3.1 | Interpretation of cline parameters

Care is needed in interpreting genomic cline parameters as there are multiple processes, including stochastic effects such as genetic drift, which can lead to deviations at individual loci from the genome-wide hybrid index (Barton, 2013; Fitzpatrick, 2013; Gompert & Buerkle, 2011). Furthermore, Gompert and Buerkle (2011) showed in simulations that two-locus epistatic Bateson-Dobzhansky-Muller (BDM) incompatibilities may not lead to steep clines (but see Gavrilets, 1997). However, an implicit rationale behind genomic cline analysis is that selection is against introgression into a whole foreign genomic background, given that locus-specific allele frequencies are regressed against whole-genome hybrid index, and this suggests multilocus rather than two-way epistasis. In a non-spatial context, it is likely that each locus with a steep cline caused by epistatic selection is in fact interacting with a subset of the whole genome, whose centre of changes in genotype frequency coincides with that of the focal locus cline centre. On the other hand, genomic cline analysis may often involve spatially distributed samples, and genome-wide hybrid index values may themselves be nonrandomly distributed in space. In this case, steep genomic clines may coincide not only with clines at interacting loci, but also with environmental transitions or barriers to dispersal (Barton & Hewitt, 1985; Kruuk et al., 1999). Examination of spatial variation in hybrid index, and coincidence of the centres of steep clines with geographic barriers or biotic or abiotic transitions in the environment, may therefore provide evidence for environment-dependent impacts of admixture. Given these possibilities, it is important to highlight that in many circumstances, the most extreme cline centres are not always the most interesting.

Little interest has been shown so far in loci with significantly shallow genomic clines, whereby both alleles are introgressing excessively into the foreign genomic background. As with other cline parameter values, shallow clines may reflect stochastic demographic effects. However, they may also represent some form of frequency-dependent selection, such as balancing selection or heterozygote advantage, maintaining polymorphism over a wide range of hybrid indices.

More complex stepped clines requiring more than 2 parameters for efficient estimation can occur across the whole genome when selection reaches a threshold value relative to recombination (Barton, 1983), or in local genomic regions where there are statistical associations (LD) with loci under direct selection against introgression (Stankowski et al., 2019; Westram et al., 2022). So far, these more complex models are only implemented for geographic cline analysis, currently in *hzar*. However, they remain relevant in genomic cline analysis despite these being non-spatial models, and their implementation would be a useful addition, especially to deal with asymmetric introgression across the cline centre potentially affecting accuracy of cline centre estimates (Baird & Macholan, 2012). This will be included in future developments of *gghybrid*.

3.2 | waic outperforms Bayesian *p* values for ranking loci and choosing a threshold

Use of the reported Bayesian *p* value is not recommended for choosing candidate loci of interest, given the peak in significance for intermediately steep clines (Figure 5a). Running the full model and both one-parameter models (with each of *v* or *centre* fixed at the null value) and ranking loci by ΔAIC or Δwaic for each parameter comparison is preferred, with waic performing best in simulations. Given the effects of an uneven distribution of hybrid indices (Figure 7, Figure S3), it may be appropriate to choose more generous thresholds for loci centred in sparsely populated regions of genome-wide hybrid index and to adjust the degree of confidence in conclusions accordingly. Furthermore, loci with differing ploidy should be treated separately (unless, for example, haploid loci are included as diploid homozygotes, which equalises sample sizes), because increased ploidy and hence number of non-missing alleles in the dataset increases statistical power.

Other things being equal, greater evidence for deviation from the null model equates to stronger evidence for an influence of

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selection. *gghybrid* is effective in identifying deviation from null values using Δwaic, but this does not exclude the possibility of 'false positives' with respect to mechanism, given that time since contact and stochastic processes can cause deviations from the genome-wide hybrid index at individual loci. An alternative method uses a type of outlier analysis based on squared Mahalanobis distances of loci from the null distribution to distinguish between loci under selection versus loci affected by stochastic processes only (Fitzpatrick, 2013; Gompert & Buerkle, 2011). The statistical threshold then allows for an expected general increase in cline parameter variation over time, while maintaining focus on the best-supported outliers. Given that Δwaic in *gghybrid* represents weight of evidence for non-null parameters, a quantile-based threshold could be applied to their distribution across loci, alongside some absolute threshold value. However, all methods rely on making a choice of statistical threshold and none completely rule out stochastic processes leading to extreme cline parameters. Evidence for selection may be augmented by hypothesis-testing. For example, an excess of steep clines in fertility-related genes compared to other functional categories would provide evidence for postzygotic reproductive isolation. Alternatively, coincidence of centres of steep clines with environmental transitions may support environmental selection.

3.3 | Conclusions

The use of geographic and genomic cline analysis is growing fast, particularly on large genomic datasets, and there are likely to be many more existing and upcoming datasets of all sizes suited to these methods. The R package *gghybrid* is intended to provide a highly accessible workflow for ecologists and evolutionary biologists familiar with R to incorporate genomic cline analysis into their studies.

AUTHOR CONTRIBUTIONS

All steps including conception and design of the package, coding and manuscript writing were carried out by RIB.

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CONFLICT OF INTEREST STATEMENT

The author declares no conflicts of interest.

DATA AVAILABILITY STATEMENT

The R package and all documentation, along with an example R code file and R code for all simulations, analyses and software comparisons presented here, and also two versions of an example dataset (previously published on Dryad: https://doi.org/10.5061/dryad.v6f4d) are available for download from Github (ribailey/gghybrid: R package for analysis of hybrids and hybrid zones (github.com)) and Zenodo, the latter with associated DOI: 10.5281/zenodo.3676498.

BENEFIT SHARING STATEMENT

Benefits from this project accrue through the sharing of an open access software package representing a vital component of modern genomic research, and addressing several issues with respect to flexibility and ease of use: (i) the software is platform-independent; (ii) it is designed to be memory-efficient and can equally easily be used on high-performance remote servers and on a standalone computer; (iii) it does not require any additional proprietary software for full performance; (iv) the software, documentation and manuscript are written to be accessible to all researchers with a reasonable background in use of R, and its use does not require any further specialist knowledge such as bioinformatics processing.

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